

ADITHYA PRASAD

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EDUCATION

MIT Manipal — B.Tech, Computer & Communication Engineering

Sep 2024 – May 2028 (Expected)

CGPA: 7.48 / 10 | Manipal, Karnataka

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Database Management Systems, Operating Systems, Computer Networks

EXPERIENCE

AI/ML Intern — CRID India (Climate Tech)

May 2026 – Present

Remote / Hybrid

- Design and develop AI systems supporting research across industrial decarbonisation, energy policy, climate finance, and carbon markets.
- Built executive intelligence workflows automating stakeholder profiling, meeting preparation, and organization research using LLMs, agentic workflows, and structured retrieval.
- Developing Feynman, a domain-specific research assistant trained on Indian energy-policy literature using continued pretraining, QLoRA fine-tuning, hybrid retrieval, and document intelligence pipelines.
- Engineered document-processing pipelines incorporating OCR, layout-aware parsing, metadata extraction, semantic chunking, and hybrid search for long-form policy documents.
- Contributed to production-ready backend services using Python, FastAPI, PostgreSQL, Docker, and cloud deployment workflows.

PROJECTS

Indian Energy Policy Research LLM (Feynman)

Python · Qwen2.5 · QLoRA · DAPT · BGE-M3 · BM25 · FastAPI · PostgreSQL

- Built an end-to-end domain-specific LLM pipeline for Indian energy-policy research combining continued pretraining (DAPT), QLoRA fine-tuning, and hybrid retrieval.
- Developed a document-ingestion system supporting OCR, layout-aware parsing, metadata extraction, semantic chunking, and benchmark dataset generation.
- Designed the system to outperform substantially larger general-purpose models on benchmarked Indian energy-policy tasks while remaining computationally efficient.

AI Life Sage (Personal AI Operating System)

React · Node.js · Express · PostgreSQL · Gemini · JWT

- Built a full-stack AI productivity platform integrating habit tracking, finance management, journaling, hobbies, and AI-powered insights.
- Designed secure authentication, REST APIs, PostgreSQL schema, audit logging, and AI-assisted analytics.
- Developed modular backend architecture with scalable API routing and cloud deployment.

Honeypot Anti-Scam Agent

India AI Summit · GUVI/HCL

Python · FastAPI · Gemini · Agentic AI

- Developed an autonomous AI honeypot capable of engaging scam callers through dynamic persona generation and intent-aware conversations.
- Combined LLM reasoning with NLP-based classification to achieve ~96% scam-detection accuracy.
- Selected among projects showcased at the India AI Summit with participation from 3,500+ attendees.

GitHub Repository Health Analyzer

Python · GitHub API · Groq · Agentic AI

- Built an AI-powered repository analysis platform that automatically evaluates repository quality, maintainability, documentation, security, and project health.
- Implemented automated scoring and natural-language reporting using LLMs, reducing manual repository assessment effort by ~40%.

PM2.5 Air Quality Prediction using Explainable AI

Published, BHTC Conference

Python · XGBoost · SHAP · Scikit-learn

- Built an explainable machine-learning pipeline for PM2.5 prediction using atmospheric and environmental variables.
- Engineered feature-selection and model optimization workflows achieving $R^2 = 99.36$, with SHAP explainability to identify dominant environmental contributors.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL

Artificial Intelligence: Machine Learning, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), Agentic AI, LLM Fine-Tuning (QLoRA/LoRA/DAPT), Prompt Engineering, Explainable AI (SHAP)

Frameworks & Libraries: PyTorch, Hugging Face Transformers, Scikit-learn, FastAPI, Flask, Express.js, React

Data & Infrastructure: PostgreSQL, MySQL, Git, GitHub, Docker, Jenkins, Google Cloud Platform

AI Tooling: Hybrid Retrieval (BGE-M3 + BM25), OCR Pipelines, Document Parsing, Vector Search, Semantic Chunking, REST APIs

RESEARCH & PUBLICATIONS

Conference Paper – Accepted, BHTC: “ML Framework for Assessing Atmospheric & Environmental Factors Affecting Air Quality” — explainable ML framework for PM2.5 prediction achieving $R^2 = 99.36$ using XGBoost and SHAP.

Journal Paper – Ongoing: “Air Quality Monitoring & Cybersecurity using Machine Learning and AES Encryption”

LEADERSHIP & ACTIVITIES

RUGVED — Management Team

MIT Manipal

- Developed and maintained the organization's official website.
- Coordinated sponsorship outreach, securing 15+ industry sponsors.
- Managed finance and external relations within a 40+ member interdisciplinary defense-robotics team.

Professional Communities: Apple Developer Group (MIT Manipal) · Manipal Information Security Team (MIST) · Consulting Club (MIT Manipal)

CERTIFICATIONS

Machine Learning Specialization — DeepLearning.AI | Deep Learning Specialization — DeepLearning.AI